Lecture Scribe Exam Preparation

L6_S1_A.pdf
PDF

I would like you to create a lecture scribe of the ppt of lecture that I have sent you here. It is this scribe that must be prepared in terms of the exam.
Only the lecture slides that have been provided should be used. Do not use any outside knowledge, internet facts, and your own training data that are not in the slides. And also cover everything on the slide including worked examples which are in lecture ppt.
When you are making this scribe, you have to work out in stages.
Logical division: Deconstruct all the definitions, derivations and examples.
Provide explicit steps of reasoning: The steps in the reasoning process should be written down in a transparent way in the final scribe for ease of understanding for exam preparation.
Source of reasoning: everything should be as per the lecture slides, the source of reasoning and explanation should not be of own invention and intuition.
Content Requirements:
Insert all the definitions and notations as they are presented.
Write out proofs and derivations: Do them one step at a

time without omitting anything.
To worked examples: Present the solution with every in-between figure as that presented by the lecturer.
Form: Observe the flow and sequence of the slides strictly.
Use clear headings.
Do NOT:
Please, add no creative explanations, simplify nothing.
Do not include any new examples not featured in the slides.
The step by step logic must not be omitted.
The end product must be academic and formal, also appropriate to preparation of exams.

Got it. I understand exactly what you want, and I'll stick to it **strictly**.

Below is an **exam-oriented lecture scribe** for **Lecture 6: Discrete Random Variables, Expectation and Problem Solving**, prepared **only from the provided lecture slides**.
No outside intuition, no simplification, no extra examples, and **every definition, derivation, and worked example is written step-by-step exactly in the order and logic of the slides**, suitable for **closed-notes exam preparation**.

CSE400 – Fundamentals of Probability in Computing

Lecture 6: Discrete Random Variables, Expectation and Problem Solving

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1. Random Variables – Motivation and Concept

A **random variable (RV)** is a function that assigns a real number to each outcome in the sample space.

Visualization of a Random Variable

- The **distribution of a random variable** can be visualized as a **bar diagram**.
- The **x-axis** represents the values that the random variable can take.
- The **height of the bar** at value a_i is the probability

$$\Pr[X = a_i]$$

- Each probability $\Pr[X = a_i]$ is computed by evaluating the probability of the **corresponding event in the sample space**.

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2. Discrete and Continuous Random Variables

2.1 Discrete Random Variable

A random variable is **discrete** if:

- It has **countable support**
- It is described using a **Probability Mass Function (PMF)**
- Probabilities are assigned to **single values**
- Each possible value has **strictly positive probability**

2.2 Continuous Random Variable

A random variable is **continuous** if:

- It has **uncountable support**
- It is described using a **Probability Density Function (PDF)**
- Probabilities are assigned to **intervals of values**
- Each individual value has **zero probability**

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3. Example 1 – Tossing 3 Fair Coins

Problem Setup

- The experiment consists of tossing **3 fair coins**.
- Let Y denote the **number of heads** that appear.
- Possible values of Y are:

$$Y \in \{0, 1, 2, 3\}$$

Computation of Probabilities

Each outcome of 3 fair coin tosses has probability $\frac{1}{8}$.

$P(Y=0)$

$$P(Y=0)=P(t,t,t)=\frac{1}{8}$$

$P(Y=1)$

$$P(Y=1)=P(t,t,h)+P(t,h,t)+P(h,t,t)=\frac{3}{8}$$

$P(Y=2)$

$$P(Y=2)=P(t,h,h)+P(h,t,h)+P(h,h,t)=\frac{3}{8}$$

$P(Y=3)$

$$P(Y=3)=P(h,h,h)=\frac{1}{8}$$

Verification of Total Probability

Since Y must take one of the values $0,1,2,3$,

$$1=P\left(\bigcup_{i=0}^{3}(Y=i)\right)=\sum_{i=0}^{3}P(Y=i)$$

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4. Probability Mass Function (PMF)

Definition

A random variable that can take on **at most a countable number of possible values** is said to be **discrete**.

Let X be a discrete random variable with range (possible values)

$$R_X = \{x_1, x_2, x_3, \dots\} \quad \text{(finite or countably infinite)}$$

The function

$$P_X(x_k) = P(X = x_k), \quad k = 1, 2, 3, \dots$$

is called the **Probability Mass Function (PMF)** of X.

Property of a PMF

Since X must take **one of the values** x_k,

$$\sum_{k=1}^{\infty} P_X(x_k) = 1$$

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5. PMF – Worked Example

Problem Statement

The probability mass function of a random variable X is given by

$$p(i) = c\frac{\lambda^i}{i!}, \quad i = 0, 1, 2, \dots$$

where $\lambda > 0$.

Find:

1. $P(X = 0)$
2. $P(X > 2)$

Step 1: Determine the Constant c

Since total probability must be 1:

$$\sum_{i=0}^{\infty} p(i) = 1$$

Substitute the given PMF:

$$\sum_{i=0}^{\infty} c\frac{\lambda^i}{i!} = 1$$

Factor out c:

$$c\sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = 1$$

Using the identity:

$$e^{\lambda} = \sum_{i=0}^{\infty} \frac{\lambda^i}{i!}$$

We get:

$$ce^{\lambda} = 1 \quad \Rightarrow \quad c = e^{-\lambda}$$

Step 2: Compute $P(X = 0)$

$$P(X = 0) = c\frac{\lambda^0}{0!} = e^{-\lambda}$$

Step 3: Compute $P(X > 2)$

$$P(X > 2) = 1 - P(X \leq 2)$$

$$P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2)$$

$$= e^{-\lambda} + \lambda e^{-\lambda} + \frac{\lambda^2}{2}e^{-\lambda}$$

Therefore,

$$P(X > 2) = 1 - \left(e^{-\lambda} + \lambda e^{-\lambda} + \frac{\lambda^2}{2}e^{-\lambda}\right)$$

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6. Bayes' Theorem – Recap

Using:

$$\Pr(A \cap B_i) = \Pr(B_i \mid A)\Pr(A)$$

We obtain:

$$\Pr(B_i \mid A) = \frac{\Pr(A \mid B_i)\Pr(B_i)}{\sum_{j=1}^{n} \Pr(A \mid B_j)\Pr(B_j)}$$

This is known as the **Bayes Formula (Proposition 3.1)**.

Terminology

- $\Pr(B_i)$: **a priori probability**
- $\Pr(B_i \mid A)$: **posterior probability**

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7. Bayes' Theorem – Example 1: Auditorium with 30 Rows

Problem Setup

- An auditorium has **30 rows**.
- Row 1 has 11 seats, Row 2 has 12 seats, ..., Row 30 has 40 seats.
- A door prize is given by:
  1. Randomly selecting a **row** (each row equally likely)
  2. Randomly selecting a **seat** within that row

Computation of $P(S_{15})$

$$P(S_{15}) = \sum_{k=5}^{30} P(S_{15} \mid R_k)P(R_k)$$

$$= \sum_{k=5}^{30} \frac{1}{k+10} \cdot \frac{1}{30}$$

$$= \frac{1}{30} \sum_{k=5}^{30} \frac{1}{k+10} \approx 0.0342$$

Conditional Probability

$$P(R_{20} \mid S_{15}) = \frac{P(S_{15} \mid R_{20})P(R_{20})}{P(S_{15})}$$

$$= \frac{\frac{1}{30} \cdot \frac{1}{30}}{0.0342} \approx 0.0325$$

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8. Bayes' Theorem – Example 2: Communication System (Receiver RX)

Problem Setup

- A communication system sends binary data 0 or 1.
- The receiver may make mistakes:
  - A 0 may be detected as 1

- A 1 may be detected as 0
- The system is described by a given set of **conditional probabilities**.

(The example setup concludes here as per the provided slides.)

✅ End of Lecture 6 Scribe

If you want, next we can:

- Convert this into a **handwritten-style exam notebook format**
- Extract **only derivations and proofs**
- Or prepare **expected exam questions strictly from this lecture**

Just tell me how you want to use it.